

Delivery Delay Prediction using Machine Learning

A machine learning system built to predict delivery delays for a logistics operation, developed during a four-week AI & ML internship at Femtosoft Technologies.

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Project Snapshot

DATASET SIZE	25,000 delivery records
INTERNSHIP DURATION	4 weeks (27 May – 27 Jun 2026)
MODELS EVALUATED	3 (Logistic Regression, Decision Tree, Random Forest)
BEST-PERFORMING MODEL	Logistic Regression
FINAL ACCURACY	89.66% (F1: 80.64%)
DASHBOARD TOOL	Microsoft Power BI

Project Overview

Logistics operators face measurable financial and customer-satisfaction costs when deliveries are delayed. This project builds a data-driven system to predict, ahead of dispatch, whether a given delivery is likely to be delayed — using operational and environmental attributes such as weather, distance, delivery partner, and vehicle type.

The work covers the complete lifecycle: data validation, exploratory analysis, feature engineering, model development, and business reporting through an interactive dashboard.

Problem Statement

Delayed deliveries reduce customer trust and increase operational cost. Identifying at-risk deliveries in advance allows a logistics team to reroute shipments, reallocate resources, or proactively notify customers before a delay occurs.

Objectives

- Analyze historical delivery data to identify the operational factors driving delays
- Develop and compare machine learning classification models for delay prediction
- Translate model output into a business-facing dashboard and set of recommendations

Technology Stack

Languages & Libraries: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Reporting: Microsoft Power BI

01 Project Methodology

Project Workflow

1. Business Understanding & Data Collection Defined the business problem and sourced a 25,000-record delivery dataset from Kaggle.	2. Data Preparation & EDA Cleaned and validated the dataset; analyzed the factors behind delivery delays.	3. Model Development Selected features and trained three classification models for comparison.	4. Reporting & Visualization Built an interactive Power BI dashboard and compiled business recommendations.
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Dataset Summary

- 25,000 delivery records, 15 columns, sourced from Kaggle
- Attributes: delivery partner, package type, vehicle type, delivery mode, region, weather condition, distance, weight, cost
- Target variable: delayed (yes / no)

Data Preprocessing

- Verified zero duplicate and zero missing records across all 15 columns
- Validated numeric ranges and categorical label consistency
- Removed two constant placeholder columns with no analytical value

Exploratory Data Analysis

Delay patterns were examined across weather conditions, delivery distance, delivery partner, and vehicle type to identify which operational factors most strongly influence delay likelihood.

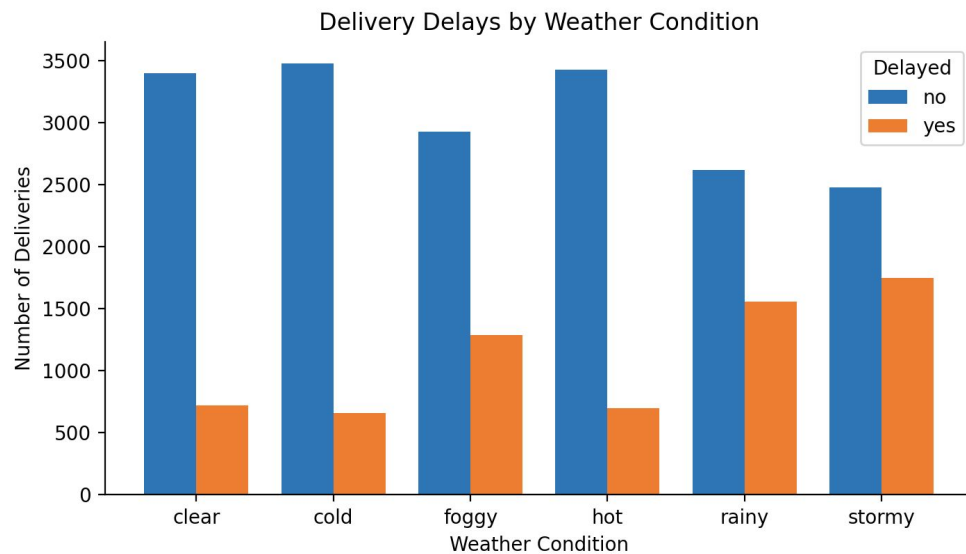


Figure 1 — Delivery delays by weather condition.

Finding: Stormy (41.45%) and rainy (37.35%) conditions recorded the highest delay rates, compared with clear (17.43%) and cold (16.02%) conditions — making weather one of the strongest predictive signals.

02 Feature Engineering & Model Development

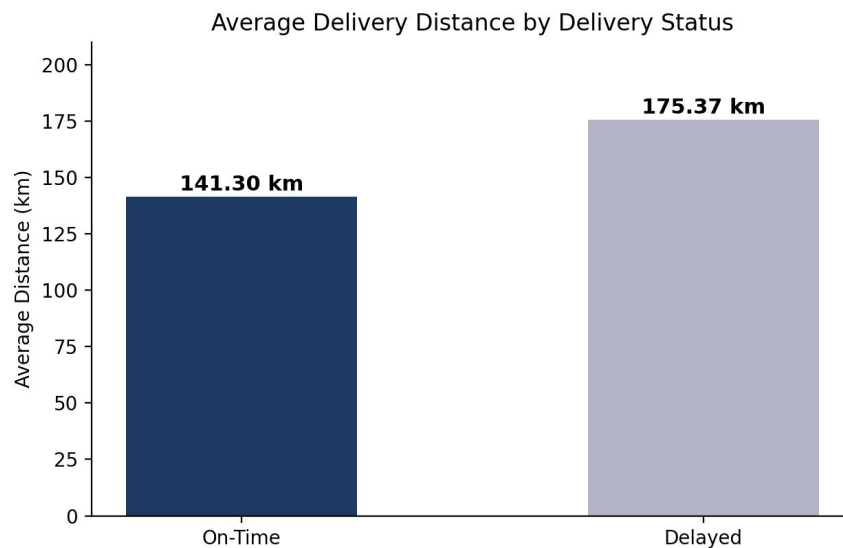


Figure 2 — Average delivery distance by delivery status.

Finding: Delayed deliveries averaged 175.37 km versus 141.30 km for on-time deliveries — a difference of roughly 34 km. Vehicle type, by contrast, showed only a minor effect, with delay rates ranging narrowly from 26.14% to 27.04% across all six vehicle categories.

Feature Selection

Nine features were selected based on availability prior to dispatch: delivery partner, package type, vehicle type, delivery mode, region, weather condition, distance, package weight, and delivery cost. Fields available only after delivery completion — delivery status and delivery rating — were excluded to prevent data leakage. After one-hot encoding of categorical variables, the dataset expanded to 36 model-ready features.

Model Training

The dataset was split 80:20 into training and test sets (20,000 / 5,000 records). Three supervised classification algorithms were trained and evaluated on the held-out test set: Logistic Regression, Decision Tree, and Random Forest.

Model Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	89.66%	80.55%	80.73%	80.64%
Decision Tree	89.38%	79.63%	80.88%	80.25%
Random Forest	89.60%	83.42%	76.16%	79.62%

Why Logistic Regression: it achieved the highest overall accuracy and F1-score, giving the most balanced trade-off between precision and recall. Decision Tree achieved marginally higher recall (80.88%) and Random Forest achieved higher precision (83.42%), but each did so at the expense of the other metric — making Logistic Regression the more consistent and business-reliable choice.

SELECTED MODEL	ACCURACY	F1 SCORE
Logistic Regression	89.66%	80.64%

03 Results & Business Recommendations

Key Findings

- Weather is a major delay driver — stormy and rainy conditions delay far more often than clear or cold conditions
- Longer delivery distances correlate strongly with delay likelihood (+34 km on average)
- Delay rates vary meaningfully across the nine delivery partners in the dataset
- Vehicle type has minimal effect on delay likelihood relative to weather and distance

Business Recommendations

- Implement weather-aware route planning and scheduling
- Prioritize monitoring of long-distance shipments
- Establish proactive delay notifications to customers
- Deploy the model to flag high-risk deliveries prior to dispatch

Delivery Outcome Distribution

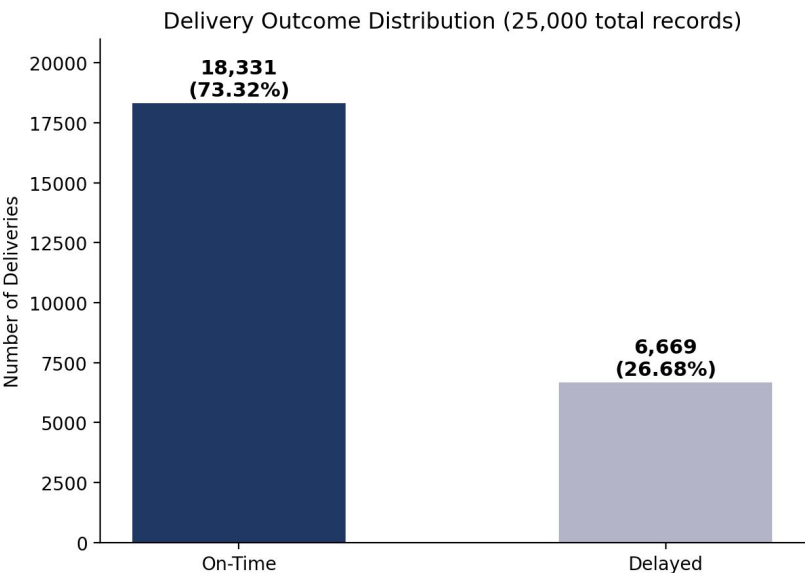


Figure 3 — Distribution of delayed and on-time deliveries (25,000 total records).

Of 25,000 deliveries analyzed, 18,331 (73.32%) were completed on time and 6,669 (26.68%) were delayed — indicating that roughly one in four deliveries is at risk without intervention.

04 Power BI Dashboard

Analytical results and model outputs were consolidated into an interactive Power BI dashboard, enabling logistics stakeholders to monitor delivery performance and identify delay patterns without direct access to the underlying dataset.

25K Total Deliveries	7K Delayed Deliveries	26.68% Delay Percentage	150.39 Average Distance	864.94 Average Cost
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Delivery Delay Prediction Dashboard

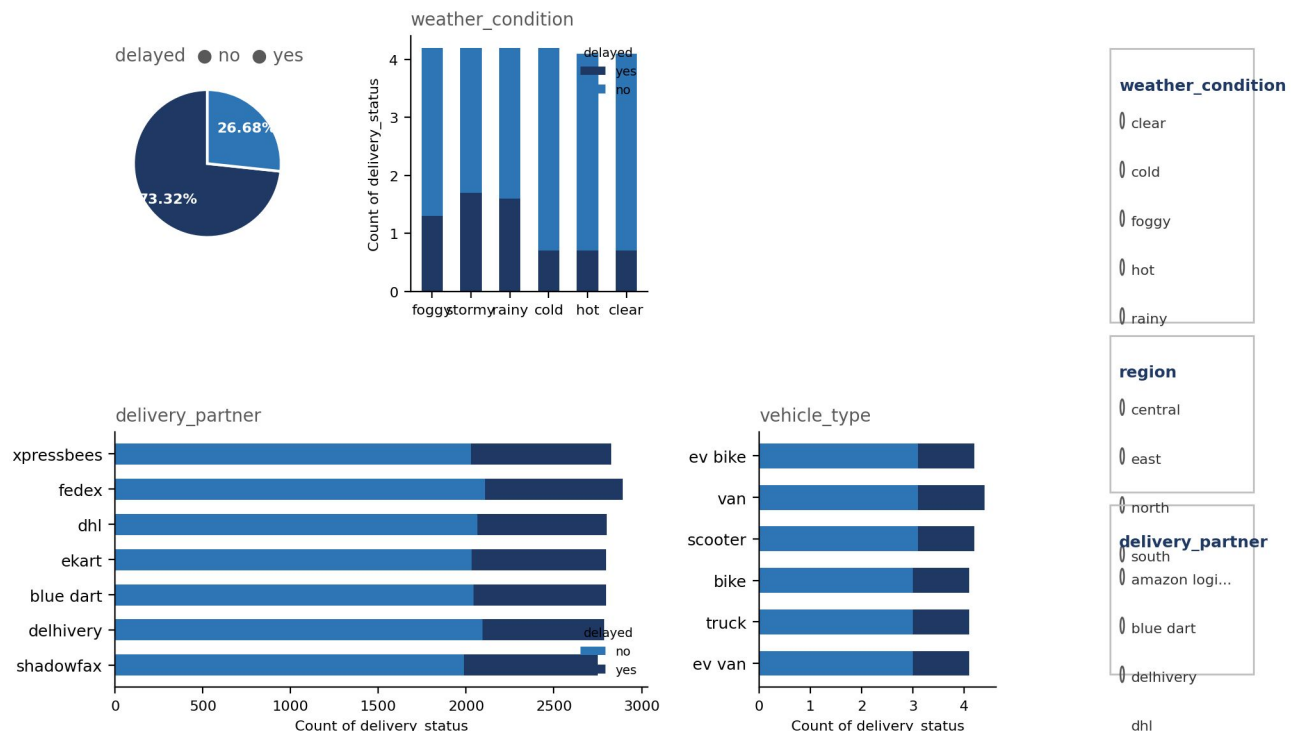


Figure 4 — Delivery Delay Prediction dashboard: KPIs, delay distribution, weather impact, delivery-partner performance, and vehicle-type breakdown.

Dashboard Features

- Five headline KPIs: total deliveries, delayed deliveries, delay percentage, average distance, average cost
- Delay distribution view and weather-impact analysis
- Delivery-partner performance comparison across all nine logistics providers
- Vehicle-type breakdown of delay rates
- Interactive filters for weather condition, region, and delivery partner

05 My Contributions

This project was completed independently as the sole intern assigned to it, covering the full data science lifecycle from raw data to business-facing output:

- Developed the end-to-end delivery delay prediction pipeline independently, from raw data ingestion through business reporting

- Validated and cleaned a 25,000-record delivery dataset, confirming zero duplicate or missing records prior to modeling
- Evaluated delay patterns across weather, distance, delivery partner, and vehicle type through exploratory data analysis
- Engineered nine predictive features and one-hot encoded categorical variables into 36 model-ready inputs
- Built and compared three classification models — Logistic Regression, Decision Tree, and Random Forest — using accuracy, precision, recall, and F1-score
- Implemented Logistic Regression as the final model based on its overall accuracy and balanced precision–recall trade-off
- Designed an interactive Power BI dashboard to communicate KPIs and delay trends to non-technical stakeholders

06 Skills, Learnings & Future Scope

Technical Skills Gained

- Data cleaning and validation
- Exploratory data analysis
- Feature engineering
- Classification model development and evaluation
- Business intelligence dashboarding

Challenges Addressed

- Prevented data leakage by excluding post-delivery fields from training
- Identified and removed non-informative placeholder columns
- Compared models beyond raw accuracy using precision, recall, and F1-score

Key Learnings

- Structured, milestone-driven execution of a multi-week analytics project
- Translating machine learning output into actionable business recommendations
- Designing dashboards for non-technical stakeholders

Future Scope

- Deploy the model as a real-time prediction service integrated with order management
- Incorporate live traffic and weather data as additional features
- Evaluate gradient boosting methods (XGBoost, LightGBM) and SHAP-based explainability